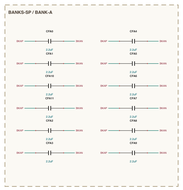
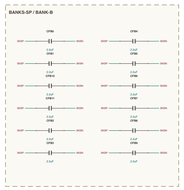
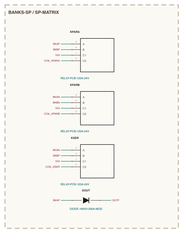
BANKS-SP / BANK-A



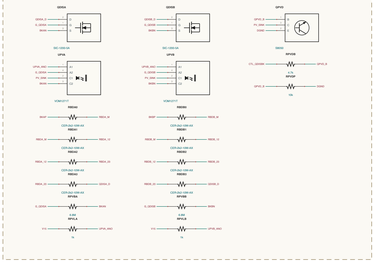
BANKS-SP / BANK-B



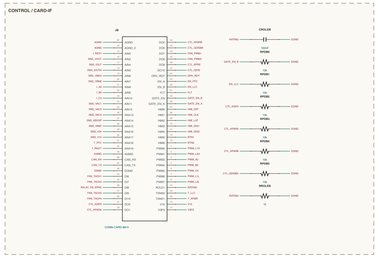
BANKS-SP / SP-MATRIX



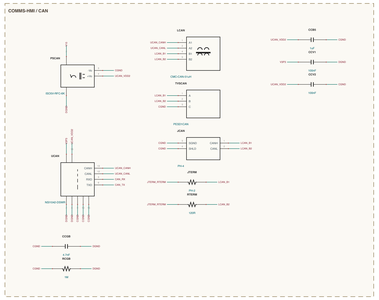
BANKS-SP / BLEEDERS



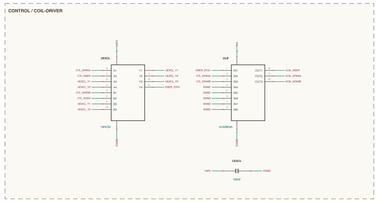
CONTROL / CARD-IF



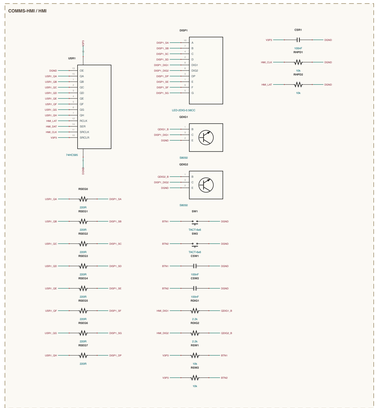
COMMS-HMI / CAN



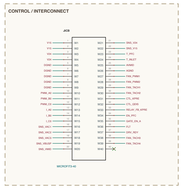
CONTROL / COIL-DRIVER



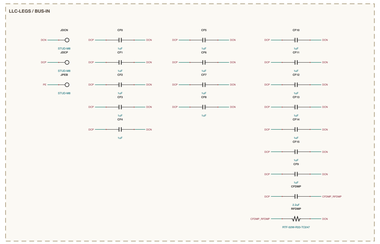
COMMS-HMI / HMI



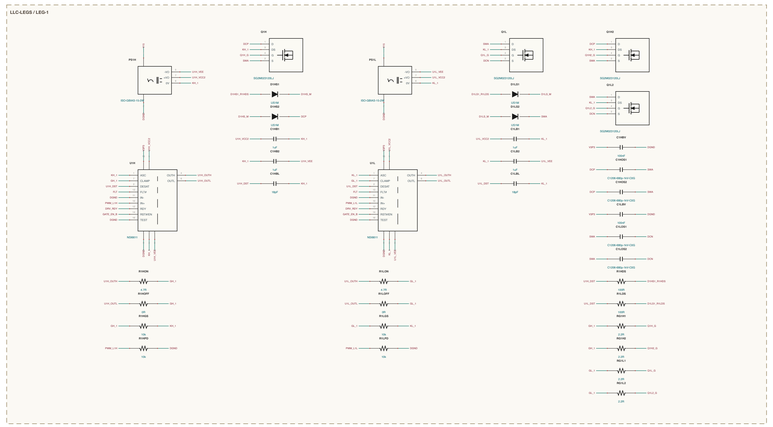
CONTROL / INTERCONNECT



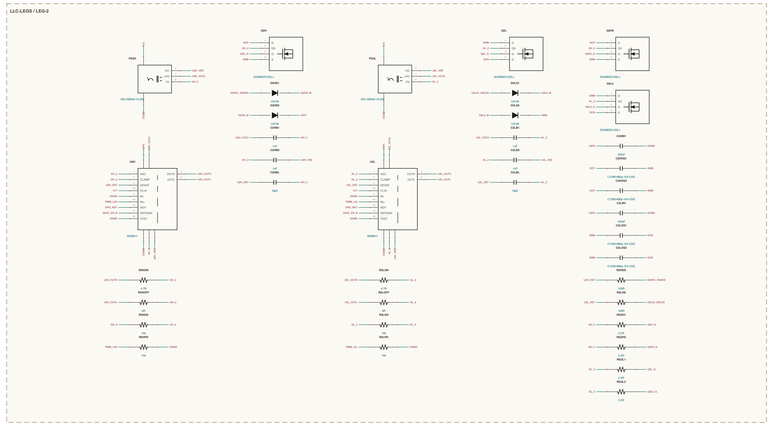
LLC-LEGS / BUS-IN



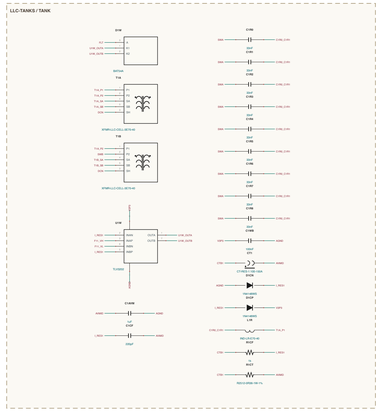
LLC-LEGS / LEG-1



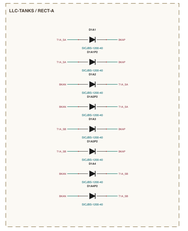
LLC-LEGS / LEG-2



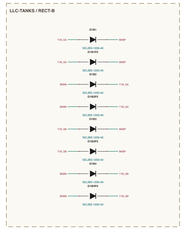
LLC-TANKS / TANK



LLC-TANKS / RECT-A



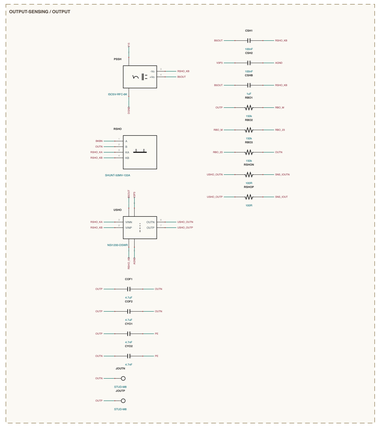
LLC-TANKS / RECT-B



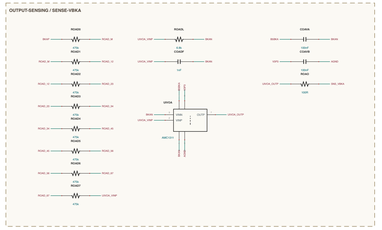
LLC-TANKS / F11-WINDOW



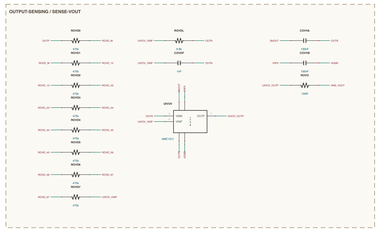
OUTPUT-SENSING / OUTPUT



OUTPUT-SENSING / SENSE-VBKA



OUTPUT-SENSING / SENSE-VOUT



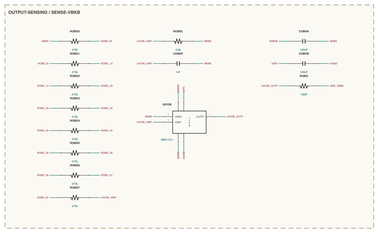
SHEET INDEX

65 kW DC-DC (upper) - 2 of 3

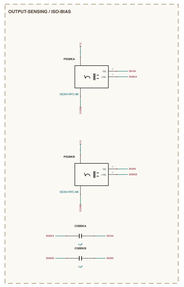
OUTPUT-SENSING - 6 sections	CONTROL - 3 sections
BANKS-SP - 4 sections	LLC-LEGS - 3 sections
LLC-TANKS - 4 sections	COMMS-HMI - 3 sections

rev D.4 - 22 sections - 306 components

OUTPUT-SENSING / SENSE-VBKB



OUTPUT-SENSING / ISO-BIAS



OUTPUT-SENSING / NTC



NET NAMING

Internal junctions are named for what they join

Lcxxfn_xPRx nods at that IC pin e.g. UFV2A_VINP

Rxxbmn_M midpoint of a series pair e.g. RBAL7A_M

Rxxbsnn_ voice-tap between Rxxbsnn1nn e.g. RV12_21

all others are explicit design nets

MAGNETICS CONSTRUCTION

Identity per docs/magnetics.md (turns, gap, litz, acceptance lines live there)

EU T14/7/8 (nxx D): 2 volts, primaries in series -> n = 0, each 3x FIT25/32 (3-stat former), 4 kV-t, pri litz 3030x0.071 * xxx 2x foil 0.08x38 tubes, Lm 21.75uH/n2 +/-7%, Bpk 150mT, BONDED

EE L-PB (nxx F): external Lr 3.95uH +/-7%, 2x ETA/3/32, 9x3, litz 1600x0.05, distributed gap, bonded

total Lr = D2 + 2x coil leakage (acceptance +/-30%) + 0.7uH loop + xx*Ls simulated, CI at 744K

bond loss screened by the ECS, bonded thermal swab (T_XFMR NTC rise at fixed load)

DC-Modules 40 kW — DC-DC board (full-bridge LLC)

DC-Modules 40 kW - board DC-DC (upper), sheet 2 of 3

2026-09-06 · rev D.4